

DRY YOUR
BEST

Strategies to improve the economic returns through dry-off management

Àlex Bach

DVM, MSc, PhD. ICREA Professor at
Department of Ruminant Production
of IRTA (Barcelona)

Luis A. Quintela Arias

DVM, MSc, PhD. Professor Animal Pathology
Department, Veterinary Faculty in Lugo,
University of Santiago de Compostela (Spain)

Victor E. Cabrera

PhD. Professor Dairy Science Department.
University of Wisconsin-Madison (USA)

Ana I. de Prado Taranilla

DVM, MSc. Global Technical Manager, Ceva
Santé Animale, Libourne (France)



Drying-off dairy cows has been widely recognized as a critical period in the production cycle of dairy cows. The strategy for drying-off cows will impact on the economy of the farm. A simulation model has been developed by the professors A. Bach (ICREA/IRTA), L. Quintela (Santiago de Compostela University) and V. Cabrera (Wisconsin-Madison University) together with Ceva Santé Animale to enable calculating the benefit of introducing the dry-off facilitator Velactis® in a given farm. This document list and explains the factors that are considered in the economical evaluation.

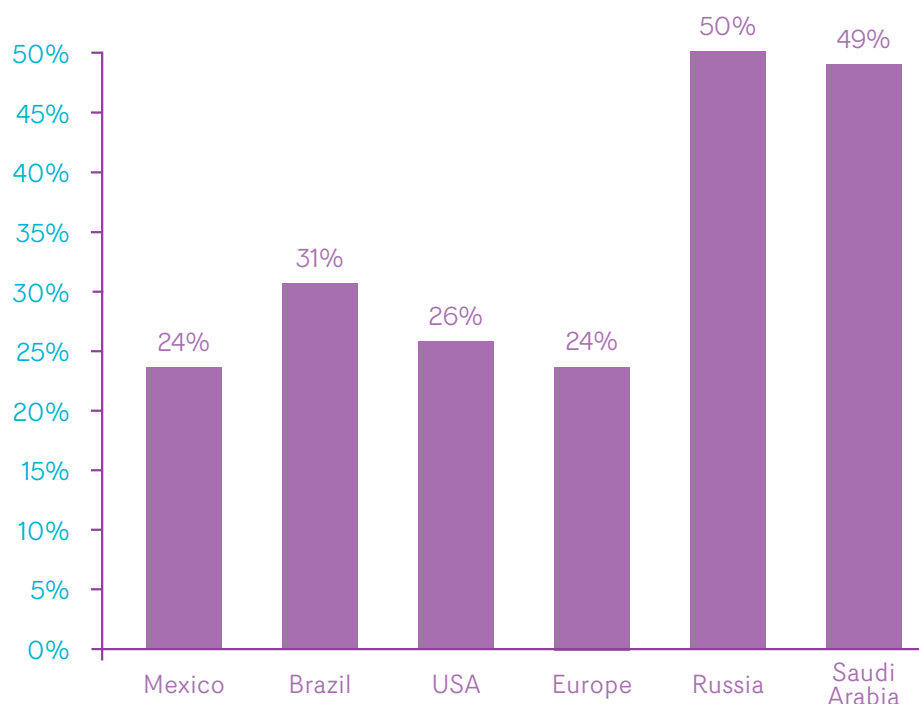
Key messages:

-  Drying off the modern high yielding dairy cow can have significant health and welfare impacts. A high yield at dry-off can result in udder engorgement. This can result in pain and thus a welfare issue. It can also result in an increase in milk leakage along an associated increase in risk of new intramammary infections.
-  Dry-off can be performed in an abrupt or gradual fashion. The type of dry-off management has consequences on the udder health, welfare and profitability of the cow.
-  Velactis together with abrupt dry-off facilitates the management at dry-off and may bring benefits in udder health, more milk sold and cow welfare. The use of Velactis will result in an economical benefit that varies depending on the type of management in the farm. In our specific examples the average net profit ranged between €42 and €60 per cow per year.

Introduction

Drying-off dairy cows has been widely recognised as a critical period in the production cycle of dairy cows (Capuco and Akers, 1999; Bachman and Schairer, 2003; Choudhary, 2014). An abrupt dry-off process (without changes in dietary or milking frequency before the dry-off) brings the benefit of maximizing the quantity of milk produced by the lactating cow. It is also a simpler process to implement for the cow and for the staff. However, it has been suggested, that with increasing milk yields over the last decades, the transition from lactating to dry-off has progressively become more challenging for cows in respect of their health and welfare (Thornton, 2010; Zobel et al., 2015). Large amounts of milk in the udder at dry-off can lead to udder engorgement and cause discomfort and pain (Bertulat et al., 2017). The high milk yield at the point of dry-off also increases the risk of new intra-mammary infections (IMI) and milk leakage (ML) (Huxley et al., 2002; Rajala-Shultz et al., 2005). The presence of ML is also associated with an increase in IMI in lactating cows. This has been demonstrated in a study of De Prado et al (2018), where quarters that experienced ML had respectively, 2.0 and 1.5 times as many odds on developing clinical mastitis (CM) or new IMI in the first 30 days after calving than cows that did not experienced ML.

The incidence of ML immediately post dry-off has been evaluated in several countries. From the combined data it may be concluded that ML post dry-off may be expected in approximately 30 % of modern dairy cows, making it a very important risk factor for subsequent new IMI.



Graph 1. Percent of cows leaking milk post dry-off in Ceva sponsored studies (adapted from de Prado et al., ADSA congress 2018).

Lowering milk production before dry-off has been found to be beneficial from the perspective of udder health and cow welfare. Several studies have examined the effects of different management strategies aimed at reducing milk production in the days before dry-off as well as accelerating the involution process early after drying-off. These include, among other strategies, dietary changes (Odensten et al., 2005; Valizaheh et al., 2008) which basically consist of curtailing nutrient supply to the cow either by feeding a low-energy diet or restricting the amount of feed available to the cows. This strategy requires two on farm actions: 1) a change in ration, and 2) a change of pens or physical location of the animal. These two actions may actually compromise immune function and thus prevent the original objective of reducing the risk of developing udder health problems. Firstly, changing animal groups will result in some social stress, and this may compromise immune function (which will be further depressed when animals are moved to a dry pen after dry-off). Second, the reduction in milk is not immediate, and animals will continue to produce milk above their nutrient supply for a few days. This milk production will then be sustained by body reserves, mainly non-esterified fatty acids, which have been linked with impaired immune function (Ster et al., 2012).

Reducing feed intake has also been linked to cows experiencing hunger (Tucker et al., 2009; Valizaheh et al., 2008), and signs of stress (Odensten et al., 2007a, b). This approach has been also proven to create a temporary state of negative energy balance that may result in metabolic problems (Odensten et al., 2005).

Cabergoline (Velactis[®], Ceva Santé Animale) is an alternative method to dry-off cows together with an abrupt dry-off (abrupt cessation of milking without any intended variation in the milk production by either feed restriction or reduced milking frequency). This dry-off facilitator lowers milk yield by reducing prolactin hormone and consequently facilitates the dry-off. As a result of applying Velactis at dry-off after the last milking, udder engorgement and ML incidence are reduced (Bach et al., 2015; Bertulat, et al., 2017) the mammary gland involution is hastened (Boutinaud et al., 2016, 2017) and consequently, the risk of new IML and the udder pain are reduced. In addition to this, Velactis will have an economic benefit by increasing the milk sales during the last days before the dry-off when compared with a gradual dry-off.

In order to demonstrate the economic benefit with the use of Velactis at dry-off in farms, Ceva Santé Animale together with recognised experts have developed a simulation model (a calculator) to be used at farm level.

Strategies to improve the economic benefit through dry-off management

In order to evaluate the most profitable changes that can be applied in a farm, firstly it is important to measure the current udder health cost that the farm has. The model calculates this cost based on the farm data and how it is managed. Secondly, the model will estimate the net benefit and the return of investment per cow after introducing Velactis. The increase in profitability may come from udder health savings and/or from getting more milk the days before the dry-off.

For some of the calculations all the milk produced from all the cows in the herd during one year has been considered. Based on this, the tool selects production level of the herd (e.g., 10,500 kg/cow per year) and the lactation curves of 1st, or $\geq 2^{\text{nd}}$ that best represent that level of production. Lactation curves have been modelled based on millions of lactation records that were used to fit a pre-defined lactation curve function, MilkBot (Ehrlich, 2011). Normalized curves represent a herd level of production. These curves are then used to calculate the expected milk production at different moments during lactation and therefore, how much milk cows will lose or gain when mastitis cases happen after calving, or changes on dry-off management respectively.

For a better understanding of the simulation model we will visit the parameters that influence the cost in udder health and the losses the farm may have due to different dry-off strategies. The calculation will be demonstrated on two different farms. The main data from these two farms are listed in **table 1**.



	Farm A	Farm B
Number of multiparous lactating cows (lactating + dry)	270	1,500
Number of primiparous lactating cows (lactating + dry)	126	600
Average 305 days milk production (kg/cow/year)	11,000	12,417
Average milk production per cow (total) (Kg/cow/lactation)	12,650	14,200
Total culling rate (%)	20	23
Total death rate (%)	2	4
Average bulk tank somatic cell count (x 1000 cells/mL)	300	225
% clinical mastitis 1st month of lactation	8	10
% milk leakage after dry-off	35	27
Average current dry period length (days)	62	69
Days of restricted feeding	7	6
Days of restricted milkings	2	1
Calving –conception interval (days)	150	160
Treatment plus withdrawal period duration (days)	9	9
Average diagnostic cost (€)	4	10
Average treatment cost (€)	26	18

Table 1. Parameters from two farms used to calculate the costs of mastitis. Farm A and B apply gradual dry-off. Farm A restricts the feeding for 7 days and the milking for 2 days before the dry-off Farm B restricts the feeding for 6 days and milking for 1 day before the dry-off.

Udder health costs related to the management applied at dry-off

— Clinical mastitis costs

Bovine clinical mastitis is a disease that has important consequences for farmers globally. Besides compromised cow welfare, it incurs economic losses through cost of treatment, production loss, and withdrawal of milk (Hogeveen et al., 2011). In addition, CM impairs reproductive performance, including a longer interval from calving to conception, more services per conception (Barker et al., 1998), lower conception rates (Santos et al., 2004; Lavon et al., 2011), and a higher risk of embryo loss (Chebel et al., 2004; McDougall et al., 2005).

The economic impact of CM is typically greater than expected by farmers and veterinarians and the cost may vary a lot depending on the farm, from €287 (Halasa et al., 2007) up to almost €1,500 (Heikkilä et al., 2012). Therefore, to calculate the real economic impact that mastitis cases have on farm, this must be calculated considering the size and the specific management on the studied farm as the number of lactating animals will have an impact on the final outcome.

Clinical mastitis cost per case can be divided into direct and indirect costs. Direct costs include: veterinary services, diagnostic testing, average treatment, non-saleable milk during the case and the withdrawal period after treatment, labour and death loss. Indirect costs include milk yield reduction of the current lactation after a clinical case, cost associated with premature culling and replacement, as well as a reduction in the reproductive performance.

Direct costs

Veterinary service cost

It is likely that in severe CM cases, the veterinarian will be called to attend the case. However, this cost will not be considered in other situations when mastitis case is diagnosed and treated by the farmer following the protocols previously established together with his/her vet. In case a veterinarian is working full time in the farm, it is estimated that the minimum time spent by a practitioner in the visit is 30 min (Richert et al., 2013). The cost will be given by the decision maker from the farm.

Diagnostic cost

This includes the average cost due for pathogen identification and antibiogram needed to select the correct antibiotic treatment. This cost will be determined by the producer or veterinarian based on their experience in the farm (e.g. numbers of mastitis milk samples that are sent to the laboratory every year).

Average treatment cost

All clinical mastitis cases should be treated as soon as possible to get the best results. Farms should have a protocol including the specific antibiotic to use, a non-steroid anti-inflammatory drug and the length of treatment. The decision making should be driven by the most frequent pathogen involved in the clinical cases and the expected efficacy of the different antibiotics tested.

Non-saleable milk

This refers to the cost of discarded milk from treated cows during the clinical case plus the withdrawal period of the drug used. It depends on the number of treated days and the milk yield the cow was producing before the clinical event. In order to calculate this milk loss, it is necessary to know the milk price the farmer receives per litre, and also the expected milk production at the time of the clinical event.

Labour cost

Undoubtedly, a sick cow requires longer time than usual as she should be moved to a sick pen and have suitable treatment administered. It has been evaluated that the time dedicated to a cow with clinical mastitis is about 45 minutes (throughout the entire process) (Van Soest et al., 2016). In order to evaluate this cost, it should be kept in mind that while staff are involved in this work, they cannot be dedicated to other farm activities. The cost will depend on the price of one hour of work in the farm. Thus, the average gross salary per month and the working hours per week per employee should be considered. Average official data per country could be used to predict this cost.

Death cost due to mastitis

In general, the relative risk of death due to mastitis is low (1.2) (Seegers et al., 2003). Knowing the incidence of CM and the total number of dead cows in the herd in one year for every 100 cows, the % of dead cows due to mastitis can be calculated (Dohoo et al., 2003). The cost of a death will be the replacement cost of a new animal.

Indirect costs

Milk yield reduction

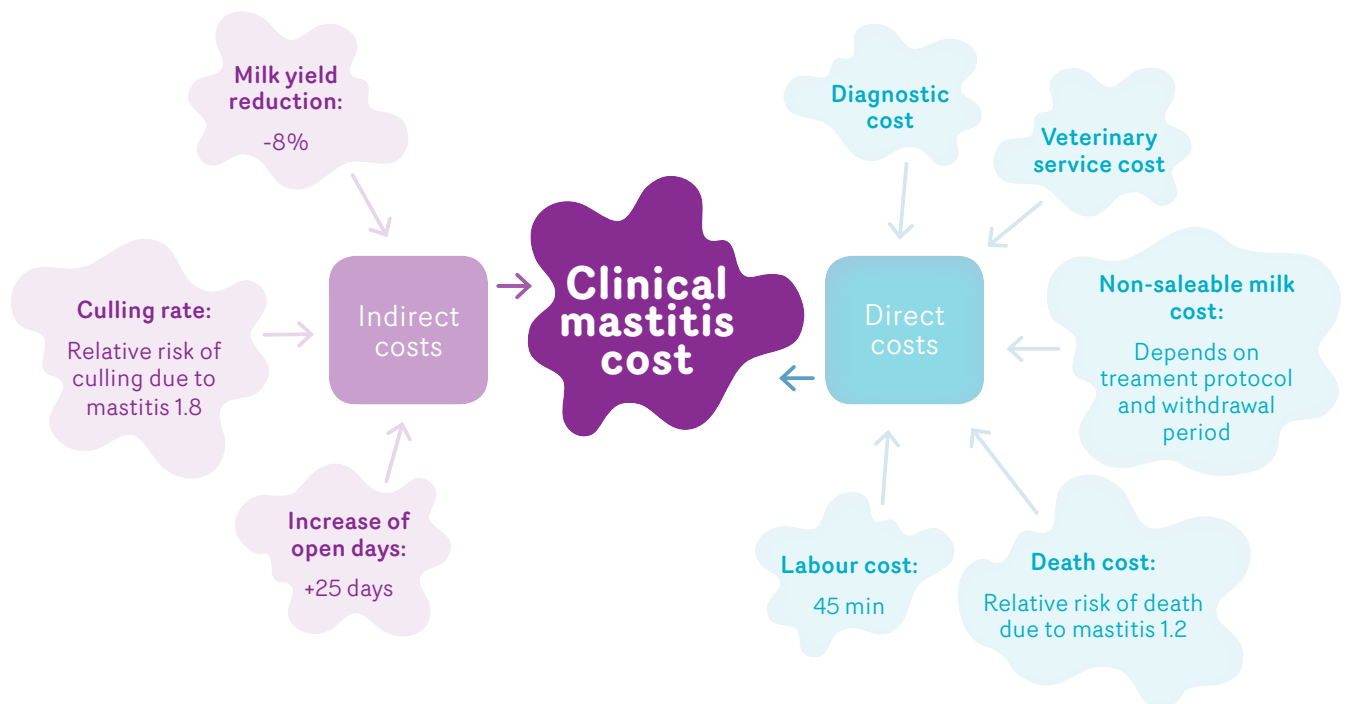
The reduction in milk production represents one of the largest percentages of economic losses. Cows will lose 8% of their expected milk production during the whole lactation when a clinical case occurs during the first month after calving (Huips et al., 2008). Therefore, the cost will be linked with the number of clinical cases at the beginning of lactation.

Culling rate

Clinical mastitis also has a cost associated with premature culling and replacement of the cow. The relative risk of culling (sold for slaughter) of a cow after suffering CM is 1.8 greater than a healthy cow (Seegers et al., 2003). Considering the incidence of CM during the first month of lactation in the farm and the total number of culled cows in the herd in one year for every 100 cows, the % of culled cows due to mastitis can be calculated (Dohoo et al., 2003). The culling cost will be the result of subtracting from the replacement cost, the residual cost of the slaughtered animal.

Reproductive performance reduction

The economic impact of CM in the first 30 days of lactation on reproductive performance was modeled using the assumption that a case of mastitis would increase by 25 the number of days between calving and conception (the service at which cows become pregnant) (Santos et al., 2004). The cost of lengthening the interval between calvings is due to extension of the final part of lactation and a reduction in the percentage of births per year. The model considers the expected milk production and the corresponding feed consumption based on the days in milk for the studied farm.



Graph 2. Clinical mastitis cost

— Subclinical mastitis costs

Subclinical mastitis is not usually treated during lactation, therefore direct costs do not exist and only indirect costs are described.

Indirect costs include: milk yield reduction and cost associated with reproductive performance reduction.

Milk yield reduction

Knowing the farm bulk tank somatic cell count (BTSCC), Van den Borne et al., (2006) assigned the percentage of lactating animals over each somatic cell count range (**Table 2**).

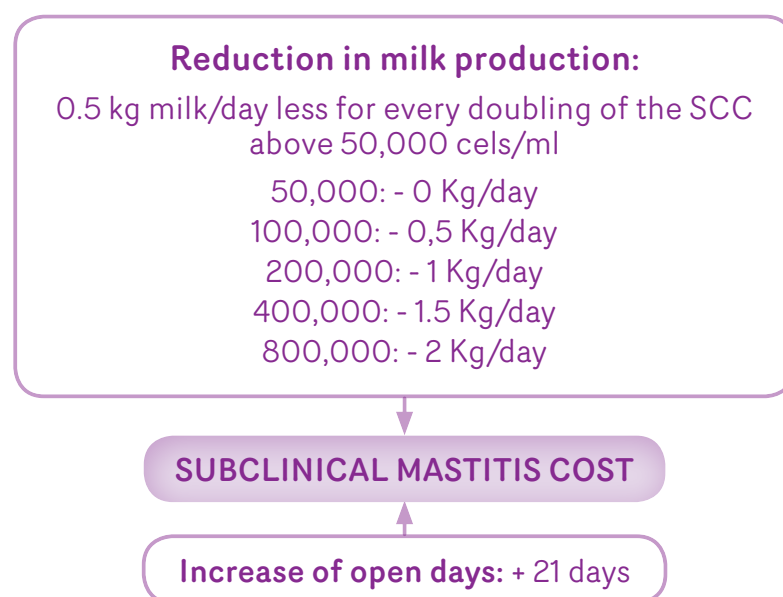
		BTSCC (x1000) (cells /mL)				
		<100	≥100/ <200	200 - 250	≥250/ <400	≥400
Individual Somatic Cell Count /mL (x1000)	0.50	80%	32%	29%	16%	0%
	50-100	16%	33%	30%	17%	0%
	100-200	3%	23%	22%	15%	3%
	200-400	1%	12%	12%	37%	31%
	>400	0%	0%	8%	15%	66%

Table 2. Distribution of the percentage of animals in each group of somatic cell count according to the BTSCC (Van der Borne et al., 2006).

Considering a cow below 50,000 cells/mL as normal, for every doubling of the SCC above 50,000 cells/ml results in a milk production loss of 0.4 kg milk/day for primiparous cows and 0.6 kg milk/day for multiparous cows (Hortet & Seegers 1998). Therefore, reducing the level of infected cows and consequently the SCC, the milk production on farm will increase (Huips et al., 2008).

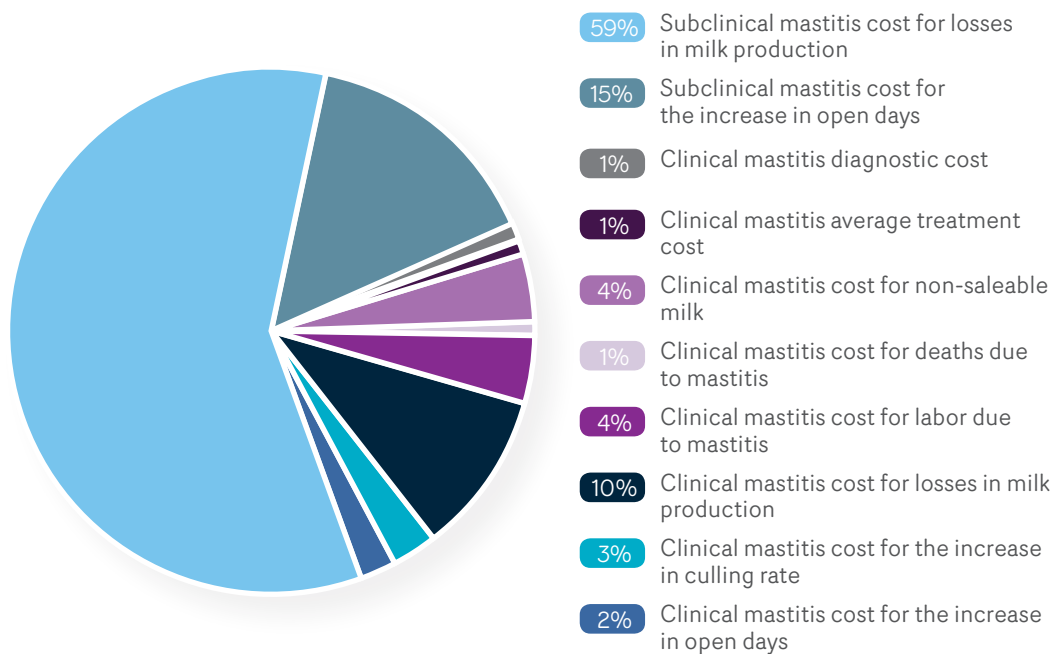
Reduction in reproductive efficacy

The economic impact of a cow with high SCC on reproductive performance was also modelled. In this case, cows with subclinical mastitis would increase by 21 the number of days between calving and conception (the service at which cows become pregnant) (Santos et al., 2004). This cost is calculated following the same calculation made in the case of clinical mastitis.

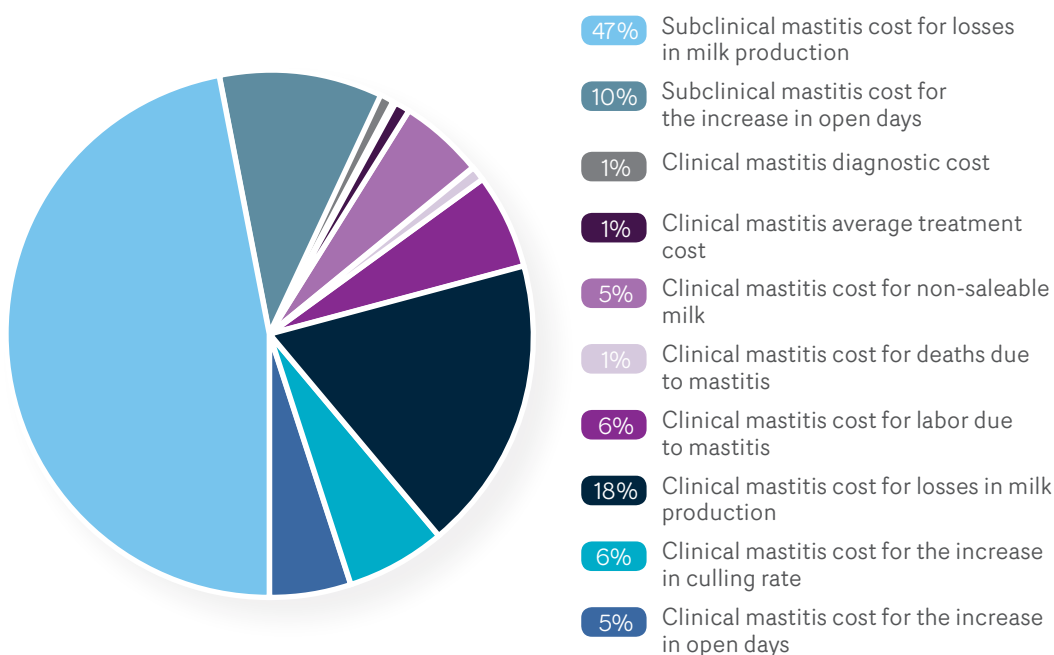


Graph 3. Subclinical mastitis cost.

Distribution mastitis costs and the value of the total cost (in euros) in farm A (graph 4) and farm B (graph 5) .



Graph 4. Distribution of mastitis costs in farm A. The total mastitis cost amounts to 85,511 euros.



Graph 5. Distribution of mastitis costs in farm B. The total mastitis cost amounts to 377,539 euros.

	Farm A	Farm B
Clinical mastitis cost (€) (% total cost)	23,031 (27%)	168,611 (45%)
Subclinical mastitis cost (€) (% total cost)	62,480 (73%)	208,928 (55%)
Total udder health cost (€)	85,511	377,539
Cost/cow (€)	216	180

Table 3. Clinical and subclinical and total mastitis cost in farm A and farm B.

Cost due to gradual dry-off management

The method of lowering milk production before dry-off and facilitating the transition from lactation to dry period has economic consequences. Using the gradual method by intermittent cessation of milking has shown to decrease milk production by 22% to 47% during the last week of lactation (Oliver et al., 1990). The reduction in dry matter intake during the 7 days before drying-off, resulted in a 34% lower milk yield compared with those where abrupt cessations of lactation was used (Tucker et al., 2009). In addition to this, gradual method implies labour costs as the farmer must apply an individual cow approach. These costs are sometimes difficult to calculate but it is important to bear it in mind. Although feed costs will decrease because a lower energy dense ration or less concentrate is used, abrupt dry-off is preferred by farmers as the management is easier and they will obtain more milk per cow per lactation. Therefore, changing from gradual to abrupt dry-off will result in economic benefit to farmers.

Considering our two farms as an example, farm A reduces the feed allowance for 7 days and also the milking frequency for 2 days before dry-off. This represents 8,470 euros of economic losses (€22/cow). Farm B, reduces the feed allowance for 6 days and the milking frequency for 1 day before dry-off and the economic losses amount to 36,151 euros (€17/cow) (**table 4**).

	Farm A	Farm B
Days of restricted feeding	7	6
Days of restricted milking	2	1
Total gradual method cost (€)	8,470	36,151
Cost/cow (€)	22	17

Table 4. Total cost when gradual dry-off method is applied in farm A and farm B.

Cost due to welfare issues

Welfare is an area of increasing importance among veterinarians, farmers, consumers and dairy processors. The dry-off process can impact negatively on several welfare aspects that can result in stress, discomfort and pain. It is difficult to evaluate the economic cost related to this area, but at the time of choosing the method to dry-off cows, all these aspects need to be considered. Both, abrupt and gradual dry-off, had disadvantages.

As mentioned before, the abrupt dry-off method in cows that are still producing a considerable amount of milk at dry-off results in the accumulation of milk in the udder. This leads to udder engorgement and udder pressure that cause discomfort and pain. There is a reduction in total lying time and the average duration of lying bouts, both being used as indicators of discomfort (Leitner et al., 2007; Cyples et al., 2012). In regards to the gradual methods, cows under feed restriction may suffer from hunger as they vocalized more and experienced a temporary state of negative energy balance with signs of stress compared with cows with no imposed feed restriction (Odensten et al., 2005, 2007; Tucker et al., 2009).

Return of investment after using Velactis

— Benefit in udder health

The use of Velactis immediately after dry-off decreases the milk yield in dairy cows and improves the udder health through a reduction in the incidence of milk leakage by 81 % and the reduction of new IMI by 21% (EU/2/15/192/001-004). Fewer new IMI will result in better udder health. These actions result in savings due to a reduction in udder health costs.

— Benefit from changing from gradual to abrupt dry-off

Velactis also allows farmers to dry-off cows abruptly without reducing the feeding or milking before dry-off. This will not only result in economic benefit as milk sales before dry-off will be maximized but also will increase the comfort of the animals. Keeping cows with the same quantity and quality of food before dry-off will avoid possible metabolic problems and they will not suffer from hunger (Oedesten et al., 2005, Valizaheh et al., 2008).

— Benefit in welfare

Cows treated with Velactis together with an abrupt dry-off significantly reduced the udder pressure and signs of pain and increased the lying time during the first day after dry off compared with controls (Bach et al., 2015; Bertulat et al, 2017). In addition to this there is no need either to move cows to another place before dry-off and consequently the stress associated with this management change (e.g.: hunger) will be avoided.

— Benefits from specific management

Recombinant bovine somatotropin (rbST) is a technology that allows a litre of milk to be produced using fewer nutrients. A meta-analysis has demonstrated that cows treated with rbST increase yield milk by about 15% over cows not treated with the hormone (Dohoo et al., 2013). When farmers use rbST (in the countries where it is allowed to do it), they try to avoid the last one or two applications before dry-off in order to reduce the milk production and facilitate dry-off.

The use of Velactis allows farmers to keep these last doses and also to extend the duration of lactation when it is possible. This procedure results in more milk sold and thereby more economic benefit at the end of lactation.

— Total benefit, total net benefit and return of investment

Considering the data from each farm and the management performed at dry-off, Velactis will bring specific benefits in each farm.

-  In farms with abrupt dry-off management at dry-off, the use of Velactis will result in benefit in udder health and welfare of the cows related to udder pain and discomfort.
-  In farms with gradual dry-off method at dry-off, the use of Velactis will facilitate the management and will result in benefit in udder health, more milk sold before the dry-off and welfare of the cows as they are kept with the same diet and in the same physical place.
-  In farms in which rbST is allowed to be used the benefit derived from the quantity of milk sold before the dry-off will be increased.

In our examples, the farm A and B will get a benefit from savings in udder health and also a benefit due to changing from gradual to abrupt dry-off.

The farm B is allowed to use rbST thus in case the farmer decides to use it, he/she will raise the benefit due to increase the milk production and extend the lactation. The manager of the farm B decides to use 2 doses more of rbST. Therefore, the farm gets 15% of more milk during the last month of lactation. The benefit derived from the rbST use, takes the price of this hormone into account.

Lastly, after considering the price of Velactis per cow (€ 25), farmers can get the net benefit after using Velactis and the ratio between the net profit and cost of investment (return of investment – ROI). A high ROI means the investment's gains favorably to its cost (**table 5**).

	Farm A	Farm B
Benefit udder health (€)	18,105	75,129
Benefit for changing from gradual to abrupt dry-off (€)	8,470	36,151
Benefit for rbST use before dry-off (€)	x	45,753
Benefit for extending lactation (€)	x	20,764
Total net benefit (€)	16,674	125,298
Net benefit / cow /year (€)	42	60

Table 5. Distribution of each benefit, total net benefit and benefit per cow per year in farm A and B.



Conclusions

The increasing milk production of the modern dairy cow is resulting in important welfare and udder health issues linked to the high level of milk production at dry off. Emphasis on reducing milk yield at drying off to prevent ML is an essential part of modern herd health management to reduce new IMI infections. The method of lowering milk production the days before dry-off and facilitating the transition from lactation to dry period has economic consequences. In order to evaluate the economic benefit of the use of Velactis on a specific farm, a mathematical model has been developed. Using the simulation model with the data from two real farms, the average net profit of applying Velactis in conjunction with the abrupt cessation of milking was between €42 and €60 per cow per year. This number includes clinical and subclinical mastitis costs, Velactis and feed costs, and milk returns at the end of lactation. Results showed that the application of Velactis immediately after the abrupt cessation of lactation is economically beneficial on the tested farms. In addition to this, Velactis facilitates management in the farm and improves cow welfare.

References

- Bach A, De-Prado A, Aris A. The effects of cabergoline administration at dry-off of lactating cows on udder engorgement, milk leakages, and lying behavior. *J Dairy Sci.* 2015 Oct;98(10):7097-101.
- Bachman, K.C., and M.L. Schairer.. Invited review: Bovine studies on optimal lengths of dry periods. *J. Dairy Sci.* 2003. 86:3027-3037. doi:10.3168/jds.S0022-0302(03)73902-2.
- Barker AR1, Schrick FN, Lewis MJ, Dowlen HH, Oliver SP. Influence of clinical mastitis during early lactation on reproductive performance of Jersey cows. *J Dairy Sci.* 1998 May;81(5):1285-90
- Bertulat S, Isaka N, de Prado A, Lopez A, Hetreau T, Heuwieser W. Effect of a single injection of cabergoline at dry off on udder characteristics in high-yielding dairy cows. *J Dairy Sci.* 2017Apr; 100(4):3220-3232.
- Boutinaud M, Isaka N, Lollivier V, Dessauge F, Gandemer E, Lamberton P, De Prado Taranilla AI, Deflandre A, Sordillo LM. Cabergotone inhibits prolactin secretion and acceterates involution in dairy cows after dry-off. *J Dairy Sci.* 2016 Jul;99(7):5707-5718.
- Boutinaud M, Isaka N, Gandemer E, Lamberton P, Wiart S, Taranilla AIP, Sordillo LM, Lollivier V. Inhibiting prolactin by cabergoline accelerates mammary gland remodeling during the early dry period in dairy cows. *J. Dairy Sci.* 2017 Dec; 100(12):9787-9798.
- Capuco, A. V, and R.M. Akers. Mammary involution in dairy animals.. *J. Mammary Gland Biol. Neoplasia* 1999. 4:137-44. doi:10.1023/A:1018769022990.
- Chebel RC, Santos JEP, Reynolds JP, Cerri RLA, Juchem SO, Overton M. Factors affecting conception rate after artificial insemination and pregnancy loss in lactating dairy cows. *Anim Reprod Sci*; 2004.84:239-55.
- Choudhary, R.K. Mammary stem cells: expansion and animal productivity. *J. Anim. Sci. Biotechnol.* 2014. 5:1-8. doi:10.1186/2049-1891-5-36.
- Cyplcs JA, Fitzpatrick CE, Leslie KE, DeVries TJ, Haley DB and Chapinal N (2012) The effects of experimentally induced *Escherichia coli* clinical mastitis on lying behavior of dairy cows. *Journal of Dairy Science* 95: 2571-2575
- De Prado-Taranilla, A. 1., A. Veithuis, L. Bertocchi, A. Appiani, O. Becvar, J. Davidek, D.Bay, L.Jimenez, K. Krömker, i.H Paduch, S.Piepers, T. van Werven, A. Veenkamp, B. Dalez, P. Le Page, M. Holstege, Y. Schukken. Presence of milk leakage on days following dry-off increased the risk of clinical mastitis and new intramammary infections at calving in commercial dairy herds around Europe. Poster at 30th World Buiatrics Congress Sapporo 28 Aug -1 Sept 2018.
- Dohoo, I.R., Martin, S.W., Stryhn, H. *Veterinary Epidemiologic Research.* 2003. AV CInc.Charlottetown, P.E.I.
- Dohoo IR, Leslie K, DesCôteaux L, Fredeen A, Dowling P, Preston A, Shewfelt W. A meta-analysis review of the effects of recombinant bovine somatotropin. 1. Methodology and effects on production. *Can J Vet Res.* 2003 Oct;67(4):241-51.
- Ehrlich, J.L. Quantifying Shape of Lactation Curves, and Benchmark Curves for Common Dairy Breeds and Parities. *The bovine practitioner.* 2011.Vol. 45, no. 1
- Halasa, T., Huijps, K., Osteras, O., Hogeveen, H. Economic effects of bovine mastitis and mastitis management: a review. *Vet. Quest.* 2007. 29, 18-31.
- Heikkila, A.M., Nousiainen, J.I., Pyorala, S. Costs of clinical mastitis with special reference to premature culling. *J. Dairy Sci.* 2012. 95, 139-150.
- Hogeveen H., Huijps K., Lam TJ. Economic aspects of mastitis: new developments. *N Z Vet J.* 2011 Jan;59(1):16-23. doi: 10.1080/00480169.2011.547165.

- Hortet, P., & Seegers, H. Calculated milk production losses associated with elevated somatic cell counts in dairy cows: review and critical discussion. *Veterinary Research*. 1998. 29(6), 497-510.
- Huijps, K., Lam, T. J., & Hogeveen, H. Costs of mastitis: facts and perception. *Journal of Dairy Research*. 2008. 75(1), 113-120.
- Huxley, J. N., Green, M. J., Green, L. E. and Bradley, A. J. Evaluation of the efficacy of an internal teat sealer during the dry period. *J. Dairy Sci*. 2002; 85(3): 551-561.
- Lavon Y., Ezra E., Leitner G., Wolfenson D. Association of conception rate with pattern and level of somatic cell count elevation relative to time of insemination in dairy cows. *J Dairy Sci*. 2011. Sep;94(9):4538-45. doi: 10.3168/jds.2011-4293.
- Leitner G, Jacoby S, Maltz E and Silanikove N (2007) Casein hydrolysate intramammary treatment improves the comfort behavior of cows induced into dry-off *Livestock Science* 110: 292-297.
- McDougall S., Rhodes FM., Verkerk G. Pregnancy loss in dairy cattle in the Waikato region of New Zealand. *N Z Vet J*. 2005 Oct;53(5):279-87.
- Odensten, M.O., Y. Chilliard, and K. Holtenius. Effects of two different feeding strategies during dry-off on metabolism in high-yielding dairy cows. *J. Dairy Sci*. 2005. 88:2072-2082. doi:10.3168/jds.S0022-0302(05)72884-8.
- Odensten M O, Holtenius K and Waller K P. Effects of two different feeding strategies during dry-off on certain health aspects of dairy cows *Journal of Dairy Science*. 2007a. 90: 898-907.
- Odensten, M. O., B. Berglund, K.P. Waller, and K. Holtenius. Metabolism and udder health at dry-off in cows of different breeds and production levels. *J. Dairy Sci*. 2007b. 90:1417-1428.
- Oliver, S.P., Shull, E.P., Dowlen, H.H. Influence of different methods of milk cessation, on intramammary infections during the peripartum period. *Proceedings of the National Mastitis Council's International Mastitis Symposium*. Indianapolis, Madison: National Mastitis Council; 1990. p. 92-7.
- Rajala-Schultz, P., J. S. Hogan, and K. L. Smith. Short Communication: Association Between Milk Yield at Dry-Off and Probability of Intramammary Infections at Calving. *J. Dairy Sci*. 2005. 88:577-579.
- Richert, K. M. Cicconi, M.J. Gamroth, Y.H. Schukken, K.E. Stiglbauer, P.L. Ruegg. Management factors associated with veterinary usage by organic and conventional dairy farms. *J. Am. Vet. Med. Assoc*. 2013. 242 pp.1732-1743
- Santos, J. E. P., Cerri, R. L. A., Ballou, M. A., Higginbotham, G. E., & Kirk, J. H. Effect of timing of first clinical mastitis occurrence on lactational and reproductive performance of Holstein dairy cows. *Animal reproduction science*. 2004.80(1-2), 31-45.
- Schukken, Y. H., J. Vanvliet, D. Vandegeer, and F. J. Grommers. A randomized blind trial on dry cow antibiotic infusion in a low somatic cell count herd. *J. Dairy Sci*. 1993. 76:2925-2930.
- Seegers, H., Fourichon, C., Beaudeau, F. Production effects related to mastitis and mastitis economics in dairy cattle herds. *Vet. Res*. 2003. 34, 475-491.
- Ster, C., M.C. Loiselle, and P. Lacasse. Effect of postcalving serum nonesterified fatty acids concentration on the functionality of bovine immune cells. *J. Dairy Sci*. 2012. 95:708-717. doi:10.3168/jds.2011-4695.
- Tucker, C.B., Lacy-Hulbert, S.J., Webster, R.J. Effect of milking frequency and feeding level before and after dry off on dairy cattle behavior and udder characteristics. *J. Dairy Sci*. 2009. 92:3194-3203.
- Thornton, P.K. Livestock production: recent trends, future prospects. *Philos. Trans. R. Soc. Biol. Sci*. 2010. 365:2853-2867. doi:10.1098/rstb.2010.0134.
- Zobel, G., Leslie, K., Weary, D.M., Von Keyserlingk, M.A.G. Gradual cessation of milking reduces milk leakage and motivation to be milked in dairy cows at dry-off. *J. Dairy Sci*. 2013. 96:5064-5071.

Van den Borne, B. H. P., Van Schaik, G., De Jong, G., & Lam, T. J. G. M. Determination of a definition of subclinical mastitis incidence based on somatic cell counts in the Netherlands. In Proceedings of the 11th Symposium of the International Society for Veterinary Epidemiology and Economics, Cairns, 2006. Australia (pp. 609-612).

Zobel, G., D.M. Weary, K.E. Leslie, and M.A.G. von Keyserlingk. Invited review: Cessation of lactation: Effects on animal welfare. J. Dairy Sci. 2015. 98:8263-8277. doi:10.3168/jds.2015-9617.

Van Soest, F. J. S., I. M. G. A. Santman-Berends, T. J. G. M. Lam, and H. Hogeveen. Failure and preventive costs of mastitis on Dutch dairy farms. J. Dairy Sci. 2016.99:8365-8374.

Valizadeh, R., D.M. Veira, and M.A.G. von Keyserlingk. Behavioural responses by dairy cows provided two hays of contrasting quality at dry-off. Appl. Anim. Behav. Sci. 2008. 109:190-200. doi:10.1016/j.applanim.2007.03.001.

